

⚡ UNIT II — Eigenvalues, Eigenvectors & Orthogonal Transformations

Engineering Mathematics · B.Tech 1st Year · Last-Day Exam Revision Formula Sheet

📖 All Key Formulas

📊 Properties Tables

📇 25 Flashcards

⚡ Exam Tips

🚀 Quick Revision

🎯 COURSE OBJECTIVES

Understand

- Eigenvalues & eigenvectors from scratch
- Characteristic equation concept

Apply

- Diagonalization of matrices
- Cayley–Hamilton theorem

Use In Engineering

- Quadratic forms & canonical form
- Orthogonal transformations

📌 EIGENVALUES & EIGENVECTORS — CORE DEFINITIONS

Definition

For an $n \times n$ matrix A , a scalar λ is an **eigenvalue** and non-zero vector X is the corresponding **eigenvector** if:

$$AX = \lambda X$$

A — square matrix | λ — scalar (eigenvalue) | X — non-zero column vector

Equivalent Form

$$(A - \lambda I)X = 0$$

Homogeneous system — non-trivial solution exists when $\det(A - \lambda I) = 0$

Characteristic Equation

$$|A - \lambda I| = 0$$

Expand determinant $\rightarrow$ polynomial in $\lambda \rightarrow$ roots are eigenvalues

Characteristic Polynomial

For 2×2 : $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$

For 3×3 : $\lambda^3 - (\text{tr } A)\lambda^2 + (\text{sum of } 2 \times 2 \text{ minors})\lambda - \det A = 0$

📌 Steps: Find Eigenvalues

- 1 Write $A - \lambda I$ (subtract λ from diagonal elements)
- 2 Compute $\det(A - \lambda I) = 0$
- 3 Expand and simplify to get characteristic polynomial
- 4 Solve polynomial $\rightarrow$ roots are $\lambda_1, \lambda_2, \dots, \lambda_n$

📌 Steps: Find Eigenvectors

- 1 Substitute each eigenvalue λ_i into $(A - \lambda_i I)X = 0$
- 2 Form the augmented matrix $[A - \lambda_i I \mid 0]$
- 3 Row-reduce to find the solution (parametric form)
- 4 Express X in terms of free variables $\rightarrow$ eigenvector

⚡ KEY RELATIONS AT A GLANCE

CORE EQUATION

$$AX = \lambda X$$

CHAR. EQUATION

$$|A - \lambda I| = 0$$

HOMOGENEOUS SYS.

$$(A - \lambda I)X = 0$$

SUM OF EIGENVALUES

$$\lambda_1 + \lambda_2 + \dots = \text{tr}(A)$$

PRODUCT OF EIGENVALUES

$$\lambda_1 \cdot \lambda_2 \cdot \dots = \det(A)$$

FOR 2×2

$$\lambda^2 - (\text{tr } A)\lambda + \det A = 0$$



PROPERTIES OF EIGENVALUES — COMPLETE TABLE

Property	Formula / Result	Meaning / Exam Note
Sum of Eigenvalues	$\lambda_1 + \lambda_2 + \dots + \lambda_n = \text{tr}(A)$	Trace = sum of diagonal elements
Product of Eigenvalues	$\lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n = \det(A)$	If any $\lambda = 0$, then $\det(A) = 0 \rightarrow A$ is singular
Triangular / Diagonal Matrix	$\lambda_i = \text{diagonal entries } a_{ii}$	Eigenvalues are the main diagonal elements
Identity Matrix I	$\lambda = 1$ (repeated n times)	All eigenvalues equal 1
Scalar Matrix (kI)	$\lambda = k$ (repeated n times)	All eigenvalues equal the scalar k
Zero Matrix	$\lambda = 0$ (all)	All eigenvalues are zero
Inverse Matrix A^{-1}	$1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_n$	Eigenvalues of A^{-1} are reciprocals of A 's eigenvalues
Matrix Power A^m	$\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$	Raise each eigenvalue to power m
Transpose A^T	Same eigenvalues as A	Transposition doesn't change eigenvalues
kA (scalar multiple)	$k\lambda_1, k\lambda_2, \dots, k\lambda_n$	Multiply each eigenvalue by k
$A - kI$	$\lambda_1 - k, \lambda_2 - k, \dots, \lambda_n - k$	Shift each eigenvalue by $-k$
Symmetric Matrix	All eigenvalues are real	Real symmetric $\Rightarrow$ real eigenvalues always
Orthogonal Matrix	$ \lambda = 1$	Eigenvalues lie on unit circle; ± 1 if real
Idempotent ($A^2 = A$)	$\lambda = 0$ or $\lambda = 1$	Only these two values possible
Nilpotent ($A^m = 0$)	$\lambda = 0$ (all)	All eigenvalues are zero for nilpotent matrices



DIAGONALIZATION OF A MATRIX

Core Formula

$$A = P D P^{-1}$$

P — modal matrix (columns = eigenvectors) · D — diagonal matrix (eigenvalues)

$$D = P^{-1} A P$$

D has eigenvalues $\lambda_1, \lambda_2, \dots$ on its diagonal

Meaning of P and D

Matrix	Contains	Role
P	Eigenvectors as columns	Change-of-basis matrix
D	$\lambda_1, \lambda_2 \dots$ on diagonal	Diagonal form of A
P^{-1}	Inverse of P	Inverse transformation

⚠ Condition for Diagonalization

- Matrix A must have **n linearly independent eigenvectors**
- If all eigenvalues are **distinct** $\rightarrow$ always diagonalizable
- Repeated eigenvalues $\rightarrow$ check independence of eigenvectors



Steps: Diagonalize A

- Find eigenvalues from $|A - \lambda I| = 0$
- Find eigenvectors for each eigenvalue
- Form P (columns = eigenvectors)
- Compute P^{-1}
- $D = P^{-1}AP \rightarrow$ diagonal matrix

$$A^n = P D^n P^{-1} \quad \text{where} \quad D^n = \text{diag}(\lambda_1^n, \lambda_2^n, \dots, \lambda_n^n)$$

Key shortcut: Computing matrix powers via diagonalization

CAYLEY-HAMILTON THEOREM

Statement (Without Proof)

"Every square matrix satisfies its own characteristic equation."

If $p(\lambda) = 0$ is the characteristic equation, then substituting the matrix A for λ gives:

$$p(A) = 0$$

The matrix A satisfies its own characteristic polynomial

For 2×2 Matrix

Char. eqn: $\lambda^2 - (\text{tr } A)\lambda + \det A = 0$

By C-H theorem:

$$A^2 - (\text{tr } A)A + (\det A)I = 0$$

For 3×3 Matrix

Char. eqn: $\lambda^3 - p_1\lambda^2 + p_2\lambda - p_3 = 0$

By C-H theorem:

$$A^3 - p_1A^2 + p_2A - p_3I = 0$$

APPLICATIONS OF CAYLEY-HAMILTON

Finding A^{-1} (Inverse)

- 1 Find characteristic polynomial $p(\lambda)$
- 2 Apply C-H: $p(A) = 0$
- 3 For 2×2: $A^2 - (\text{tr } A)A + (\det A)I = 0$
- 4 Rearrange: $A^{-1} = [A - (\text{tr } A)I] / \det(A)$

$$A^{-1} = (1/\det A)[A - (\text{tr } A)I]$$

Valid only when $\det(A) \neq 0$

Finding Powers of A (A^n)

- 1 Get characteristic equation of A
- 2 Express higher powers in terms of lower ones using C-H
- 3 For 2×2: $A^2 = (\text{tr } A)A - (\det A)I$
- 4 Use recursion: $A^3 = A \cdot A^2$, $A^4 = A \cdot A^3$, etc.

$$A^n = \alpha A + \beta I$$

Coefficients α, β found by substituting eigenvalues

⚡ Cayley-Hamilton Quick Tips

- Always verify the characteristic equation first before applying C-H
- For finding A^{-1} : Never use C-H if the matrix might be singular ($\det = 0$)
- Shortcut: For 2×2 matrix, A^2 can always be written as $(\text{tr } A) \cdot A - (\det A) \cdot I$
- The theorem holds for all square matrices — real, complex, symmetric, etc.



QUADRATIC FORMS

Definition

A homogeneous polynomial of degree 2 in variables $x_1, x_2, \dots, x_n$

$$Q = X^T A X$$

X — column vector of variables · A — symmetric coefficient matrix

For 2 Variables (x, y)

$$Q = ax^2 + 2hxy + by^2$$

Matrix form: $A = [[a, h], [h, b]]$

For 3 Variables (x, y, z)

$$Q = ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy$$

Symmetric Matrix A:

a	h	g
h	b	f
g	f	c



NATURE OF QUADRATIC FORMS — CLASSIFICATION TABLE

Nature	Eigenvalue Condition	Q value	Index (r) / Signature	Example
Positive Definite PD	All $\lambda_i > 0$	$Q > 0$ for all $X \neq 0$	$r = n$ (all positive)	$x^2 + y^2 + z^2$
Positive Semi-definite PSD	All $\lambda_i \geq 0$ (some = 0)	$Q \geq 0$; $Q=0$ for some $X \neq 0$	$r < n$, no negative λ	$x^2 + y^2$
Negative Definite ND	All $\lambda_i < 0$	$Q < 0$ for all $X \neq 0$	$r = n$ (all negative)	$-x^2 - y^2 - z^2$
Negative Semi-definite NSD	All $\lambda_i \leq 0$ (some = 0)	$Q \leq 0$; $Q=0$ for some $X \neq 0$	$r < n$, no positive λ	$-x^2 - y^2$
Indefinite IND	Both positive & negative λ	Q takes +ve and -ve values	Mixed signs present	$x^2 - y^2$

Sylvester's Criterion (Positive Definite)

A symmetric matrix A is PD $\leftrightarrow$ all leading principal minors are positive:

$$\Delta_1 = a_{11} > 0, \Delta_2 > 0, \Delta_3 > 0, \dots$$

Index & Signature

- Index (r)** = number of positive eigenvalues
- Signature (s)** = (no. of positive λ) – (no. of negative λ)
- Rank** = number of non-zero eigenvalues



ORTHOGONAL TRANSFORMATION

Orthogonal Matrix — Definition

$$P^T P = P P^T = I$$

Columns (and rows) are orthonormal vectors

Equivalently: $P^{-1} = P^T$

Properties: $\det(P) = \pm 1$, $|\text{eigenvalues}| = 1$

Orthogonal Diagonalization

$$D = P^T A P$$

P is orthogonal ($P^T P = I$)

$$A = P D P^T$$

Since $P^{-1} = P^T$ for orthogonal matrices

Gram–Schmidt Orthogonalization (Converting eigenvectors to orthonormal)

Step 1: Start with X_1

$$u_1 = X_1$$

Step 2: Orthogonalize X_2

$$u_2 = X_2 - (X_2 \cdot u_1 / |u_1|^2) u_1$$

Step 3: Normalize

$$\hat{e}_1 = u_1 / |u_1|, \hat{e}_2 = u_2 / |u_2|$$



REDUCTION TO CANONICAL FORM

- Write the quadratic form $Q = X^T A X$
- Find eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of A
- Find normalized eigenvectors (orthonormal set)
- Form orthogonal matrix P (columns = normalized eigenvectors)
- Apply $X = P Y$ substitution
- Canonical form: $Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$

Canonical Form Result

$$Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$$

No cross-product terms — pure quadratic form

Why useful? Easier to determine nature (PD/ND/Indefinite) just by checking signs of λ_i



EXAM FLASHCARDS (25 Q&A)

Q: What is an eigenvalue? λ such that $AX = \lambda X$ has non-trivial solution $X \neq 0$ Scalar associated with eigenvector	Q: What is an eigenvector? Non-zero vector X such that $AX = \lambda X$ Only direction, not unique magnitude
Q: Characteristic equation? $ A - \lambda I = 0$ Set determinant of $(A - \lambda I)$ to zero	Q: Sum of eigenvalues? $\lambda_1 + \lambda_2 + \dots + \lambda_n = \text{tr}(A)$ Sum of diagonal elements of A
Q: Product of eigenvalues? $\lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n = \det(A)$ Determinant of the matrix	Q: Eigenvalues of triangular matrix? Diagonal elements $a_{11}, a_{22}, \dots, a_{nn}$ Read directly off the main diagonal
Q: Eigenvalues of identity matrix I? All eigenvalues = 1 $\lambda = 1$ repeated n times	Q: Eigenvalues of A^{-1}? $1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_n$ Reciprocals of A 's eigenvalues
Q: Eigenvalues of A^m? $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ Raise each eigenvalue to power m	Q: Cayley–Hamilton theorem states? Every square matrix satisfies its own characteristic equation $p(A) = 0$ Substitute A in place of λ
Q: C-H for 2×2: $p(A) = 0$ means? $A^2 - (\text{tr } A)A + (\det A)I = 0$ Used to find A^{-1} and higher powers	Q: A^{-1} using C-H (2×2)? $A^{-1} = [A - (\text{tr } A)I] / \det(A)$ Derived from C-H theorem
Q: Condition for diagonalization? n linearly independent eigenvectors Distinct λ 's always satisfy this	Q: Diagonalization formula? $A = PDP^{-1}$ and $D = P^{-1}AP$ P = modal matrix, D = diagonal
Q: What does P contain in diagonalization? Eigenvectors as columns (modal matrix) Order must match eigenvalue order in D	Q: What does D contain? Eigenvalues on the diagonal; zeros elsewhere $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$
Q: Quadratic form definition? $Q = X^TAX$ where A is symmetric Homogeneous polynomial of degree 2	Q: Positive definite condition? All eigenvalues of A are strictly positive ($\lambda_i > 0$) $Q > 0$ for all $X \neq 0$
Q: Indefinite quadratic form means? A has both positive and negative eigenvalues Q takes both +ve and -ve values	Q: Orthogonal matrix condition? $P^T P = I$ i.e., $P^{-1} = P^T$ Columns are orthonormal
Q: Orthogonal diagonalization? $D = P^TAP$, $A = PDP^T$ P is orthogonal ($P^T P = I$)	Q: Canonical form of Q? $Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$ No cross terms; $X = PY$ substitution used
Q: Rank of quadratic form? Number of non-zero eigenvalues = rank of matrix A	Q: Index of quadratic form? Number of positive eigenvalues (r) Used in classifying nature of form
Q: Signature of quadratic form? $s = (\text{no. of +ve } \lambda) - (\text{no. of -ve } \lambda)$ $s = 2r - \text{rank}$	

⚡ EXAM TIPS & COMMON MISTAKES

✗ Common Errors in Eigenvalue Calculation

- Forgetting to **subtract λ from ALL diagonal elements**, not just one
- Sign errors when expanding 3×3 determinant (cofactor expansion)
- Solving characteristic polynomial incorrectly — always verify: $\lambda_1 + \lambda_2 = \text{tr}(A)$
- Not checking: $\lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n = \det(A)$ after finding eigenvalues
- Using wrong sign in $(A - \lambda I)$ — it is A MINUS λI , not A PLUS λI

✗ Errors in Eigenvector Finding

- Not putting the system back as **homogeneous** (RHS = 0)
- Choosing zero vector as eigenvector — eigenvectors must be non-zero
- Not expressing answer in parametric form (free variable method)
- Forgetting to normalize when orthogonal transformation is asked

✗ Diagonalization Mistakes

- Putting eigenvectors in wrong order in P (must match λ order in D)
- Trying to diagonalize a non-diagonalizable matrix
- Computing P^{-1} incorrectly — always verify: $P^{-1} \cdot P = I$

⚡ Smart Shortcuts

- **2x2 char. eq:** $\lambda^2 - (\text{tr } A)\lambda + \det A = 0 \rightarrow$ use quadratic formula
- **Check:** After finding λ , verify sum = trace and product = det
- **Triangular matrix:** Eigenvalues = diagonal entries directly
- **Symmetric matrix:** All eigenvalues are real — saves computation
- **Orthogonal matrix:** Use P^T instead of computing P^{-1}
- **Idempotent ($A^2 = A$):** $\lambda = 0$ or 1 only — verify using this
- **Nilpotent ($A^m = 0$):** All eigenvalues = 0

⚡ When Diagonalization Fails

- Repeated eigenvalues with insufficient independent eigenvectors
- Geometric multiplicity < Algebraic multiplicity $\rightarrow$ not diagonalizable
- Use Jordan Normal Form for such matrices (if asked)

🚀 ULTRA-QUICK REVISION REFERENCE SHEET

⚡ ALL KEY FORMULAS — ONE PAGE

EIGENVALUE EQ.

$$AX = \lambda X$$

CHAR. EQUATION

$$|A - \lambda I| = 0$$

HOMOGENEOUS SYS.

$$(A - \lambda I)X = 0$$

SUM OF λ

$$\sum \lambda = \text{tr}(A)$$

PRODUCT OF λ

$$\prod \lambda = \det(A)$$

2x2 CHAR. POLY

$$\lambda^2 - (\text{tr } A)\lambda + \det A = 0$$

DIAGONALIZATION

$$A = PDP^{-1}$$

D MATRIX

$$D = P^{-1}AP$$

MATRIX POWER

$$A^n = PD^nP^{-1}$$

C-H THEOREM

$$p(A) = 0$$

C-H (2x2)

$$A^2 - (\text{tr } A)A + (\det A)I = 0$$

A^{-1} VIA C-H

$$A^{-1} = [A - (\text{tr } A)I] / \det A$$

QUADRATIC FORM

$$Q = X^TAX$$

CANONICAL FORM

$$Q = \sum \lambda_i y_i^2$$

ORTHOGONAL MATRIX

$$P^T P = I, P^{-1} = P^T$$

ORTH. DIAG.

$$D = P^TAP$$

POSITIVE DEFINITE

$$\text{ALL } \lambda_i > 0$$

INDEFINITE

$$\text{Mixed signs of } \lambda_i$$



COMPARISON: DIAGONALIZATION vs ORTHOGONAL TRANSFORMATION

Aspect	General Diagonalization	Orthogonal Transformation
Formula	$D = P^{-1}AP$	$D = P^TAP$
Matrix P type	Any invertible matrix of eigenvectors	Orthogonal matrix (orthonormal eigenvectors)
Condition on A	n linearly independent eigenvectors	A must be real symmetric
P^{-1}	Computed normally	$P^{-1} = P^T$
Application	Matrix powers, general diagonalization	Reducing quadratic form to canonical form